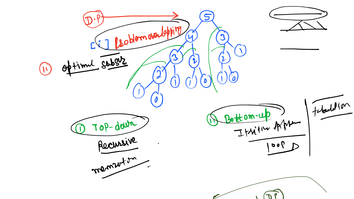
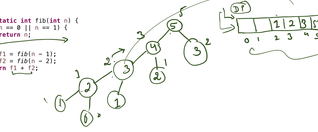
Those who cannot remember the past are condemned to repeat it.

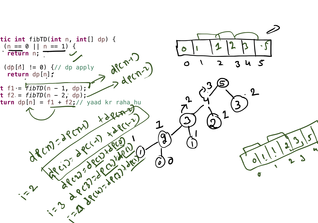
-Dynamic Programming

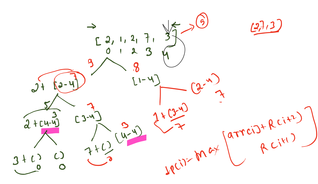



```
public static int fib(int n) {
    if (n == 0 || n == 1) {
        return n;
    }
    int f1 = fib(n - 1);
    int f2 = fib(n - 2);
    return f1 + f2;
}
```

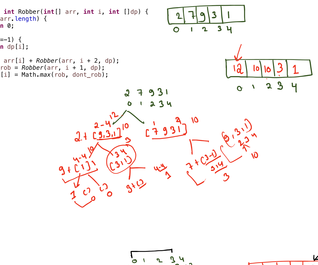


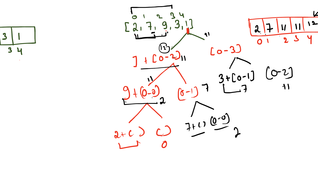

```
public static int fibTD(int n, int[] dp) {
    if (n == 0 || n == 1) {
        return n;
    }
    if (dp[n] != 0) {// dp apply
        return dp[n];
    }
    int f1 = fibTD(n - 1, dp);
    int f2 = fibTD(n - 2, dp);
    return dp[n] = f1 + f2;// yaad kr rahe ho
}
```

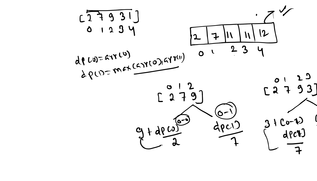


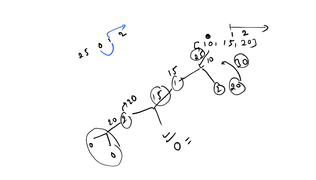


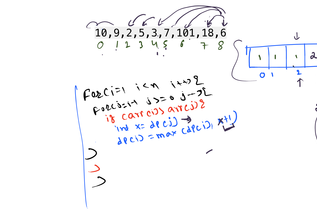

```
public static int Robber(int[] arr, int i, int []dp) {
    if (i >= arr.length) {
        return 0;
    }
    if(dp[i]!=-1) {
        return dp[i];
    }
    int rob = arr[i] + Robber(arr, i + 2, dp);
    int dont_rob = Robber(arr, i + 1, dp);
    return dp[i] = Math.max(rob, dont_rob);
}
```

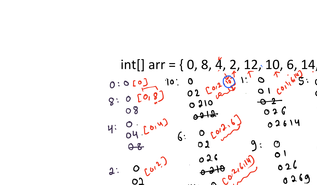


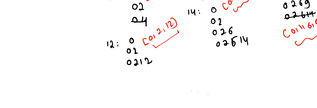













```
int[] arr = { 0, 8, 4, 2, 12, 10, 6, 14, 1, 9, 5, 13, 3, 11, 7, 15 };
```